

ELITE MATHEMATICS

Pearson Edexcel International GCSE

Modular Unit 1

Adaptive Classified

OL-T49 · Sine Rule, Cosine Rule and Area of a Triangle

Questions

Student book · questions with writing space

■ 13 pages

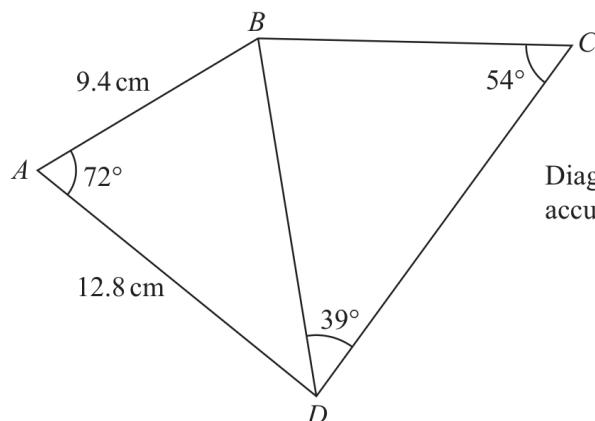
FAMILY MAP

Sine Cosine Rule Area of Triangles

F01	Resolve mixed non-right-triangle chains	6 Q 31 marks
F02	Use the sine rule	3 Q 16 marks
F03	Use the cosine rule	1 Q 3 marks
F04	Use area equals one-half $ab \sin C$	1 Q 3 marks

Working Unit 13

4MA1-1H Q20

UNIT 1**5 marks****20**Diagram **NOT**
accurately drawn

Work out the length of BC
Give your answer correct to 3 significant figures.

..... cm

(Total for Question 20 is 5 marks)

Question 13

4MA1-1H Q19

MODULAR UNIT 1

5 marks

19 Here is quadrilateral $ABCD$

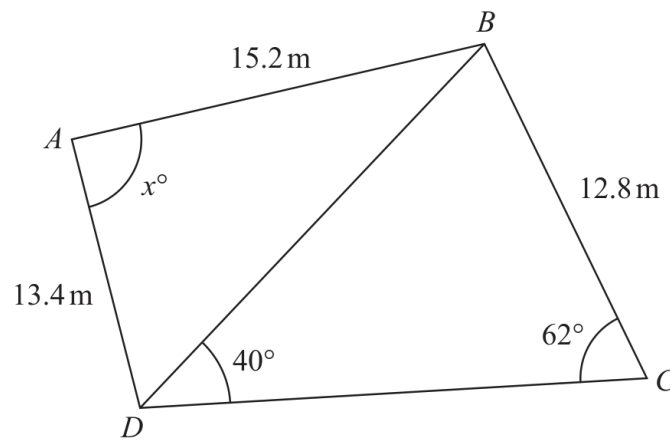


Diagram **NOT**
accurately drawn

Work out the value of x
Give your answer correct to 3 significant figures.

$x = \dots\dots\dots$

Question 09

4MA1-1H Q10 demand

MODULAR UNIT 1

5 marks

10 Here is triangle ABC

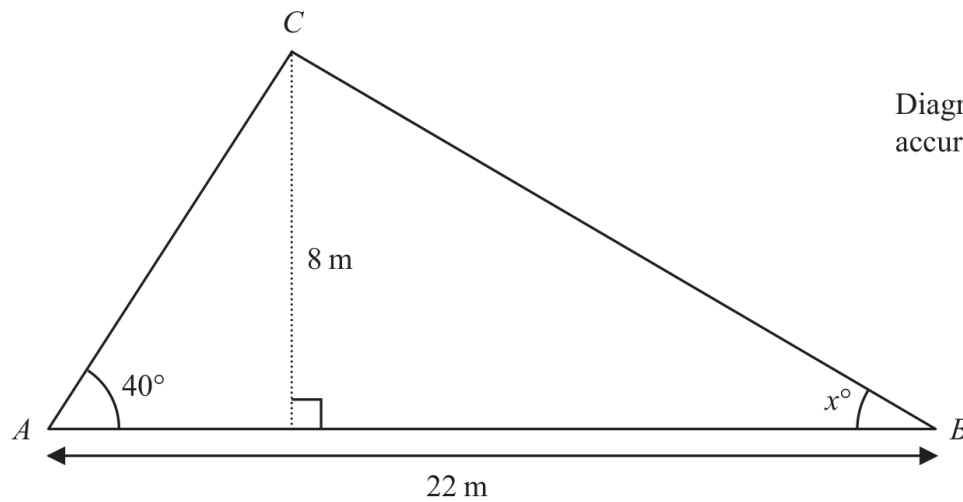


Diagram **NOT**
accurately drawn

Work out the value of x
Give your answer correct to one decimal place.
Show your working clearly.

$x =$

Question 26

4MA1-2H Q22

MODULAR UNIT 1

5 marks

22 The diagram shows a triangle ABC and a flagpole BF

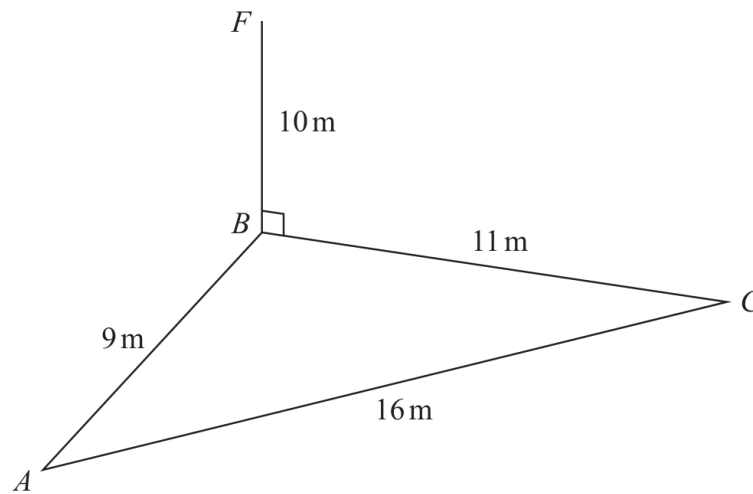


Diagram **NOT**
accurately drawn

A , B and C are points on horizontal ground.

BF is vertical.

$$AB = 9 \text{ m} \quad BC = 11 \text{ m} \quad AC = 16 \text{ m} \quad BF = 10 \text{ m}$$

D is the point on AC such that angle $BDC = 90^\circ$

Work out the size of the angle of elevation of the point F from the point D
Give your answer correct to one decimal place.

Question 12

4MA1-1H Q18 Q18

MODULAR UNIT 1

5 marks

18 Here is triangle ABC

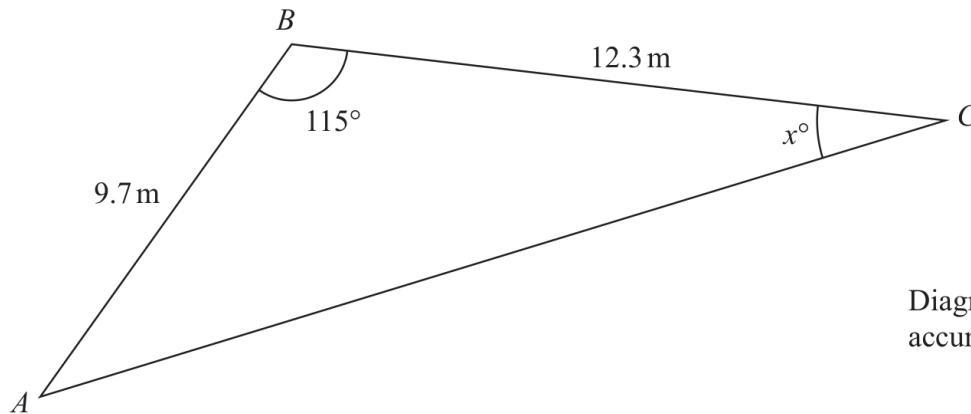


Diagram **NOT**
accurately drawn

Work out the value of x
Give your answer correct to 3 significant figures.

Question 27

4MA1-2H Q23

MODULAR UNIT 1

6 marks

23 The diagram shows triangle PQR

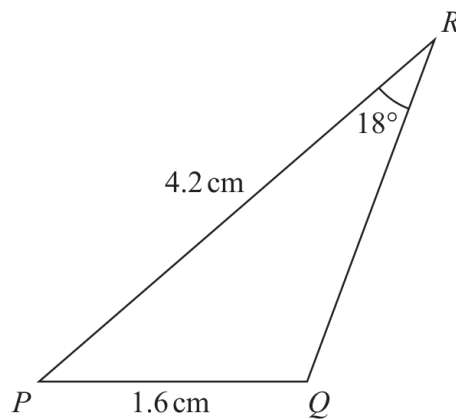


Diagram **NOT**
accurately drawn

$$PQ = 1.6 \text{ cm}$$

$$PR = 4.2 \text{ cm}$$

$$\text{Angle } PRQ = 18^\circ$$

Given that angle PQR is obtuse,

work out the area of triangle PQR

Give your answer correct to 3 significant figures.

..... cm^2

14 The diagram shows parallelogram $EFGH$.

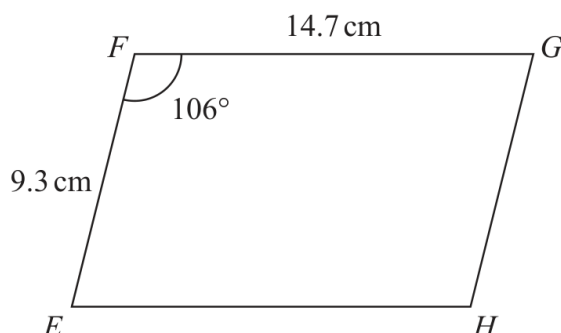


Diagram **NOT**
accurately drawn

$$EF = 9.3 \text{ cm}$$

$$FG = 14.7 \text{ cm}$$

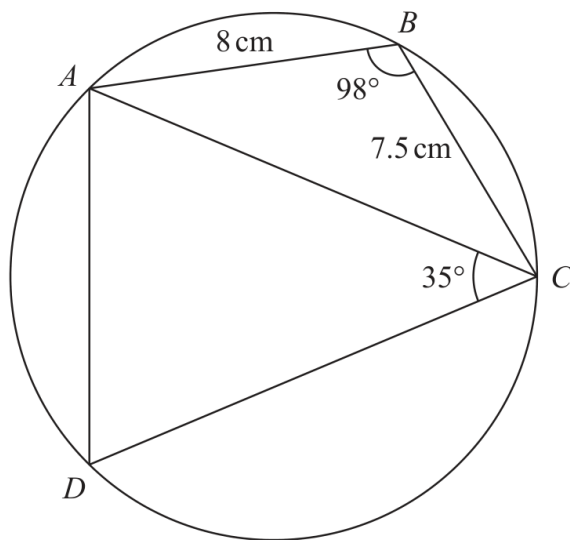
$$\text{Angle } EFG = 106^\circ$$

- (a) Work out the area of the parallelogram.
Give your answer correct to 3 significant figures.

- (b) Work out the length of the diagonal EG of the parallelogram.
Give your answer correct to 3 significant figures.

..... cm
(3)

Diagram **NOT**
accurately drawn


$$\begin{aligned} AB &= 8 \text{ cm} \\ BC &= 7.5 \text{ cm} \\ \text{Angle } ABC &= 98^\circ \\ \text{Angle } ACD &= 35^\circ \end{aligned}$$

..... cm

A quadrilateral $ABCD$ is shown with diagonal AC . The side lengths are $AB = 9\text{ cm}$, $BC = 12\text{ cm}$, and $CD = 7\text{ cm}$. The interior angles are $\angle ABC = 60^\circ$ and $\angle ACD = 84^\circ$.

Calculate the area of quadrilateral $ABCD$.
Give your answer correct to 3 significant figures.
Show your working clearly.

..... cm²

The diagram shows two triangles, $\triangle ABC$ and $\triangle EDC$, that intersect at vertex C . The segments BC and EC are collinear, as are AC and DC . The following measurements are given:

- $BC = 31$ cm
- $AC = 24$ cm
- $EC = 19$ cm
- $DC = 16$ cm
- $\angle A = 64^\circ$

 $AC = 24 \text{ cm} \quad BC = 31 \text{ cm} \quad CE = 19 \text{ cm} \quad CD = 16 \text{ cm}$

Angle $BAC = 64^\circ$

Work out the length of DE

Give your answer correct to 3 significant figures.

..... cm

(Total for Question 16 is 5 marks)

19 The diagram shows triangle ABC

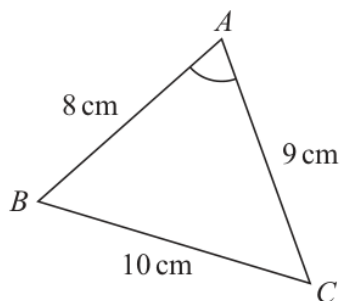


Diagram **NOT**
accurately drawn

$$AB = 8 \text{ cm} \quad BC = 10 \text{ cm} \quad CA = 9 \text{ cm}$$

Work out the size of angle BAC

Give your answer correct to one decimal place.

(Total for Question 19 is 3 marks)